

BUILDING AND DEPLOYING APPLICATIONS WITH DOCKER

1. Overview: Three-Tier Architecture

A Three-Tier Application consists of:

- **Frontend (React)** – User Interface
- **Backend (Node.js)** – Application logic and API
- **Database (MySQL)** – Data persistence layer

Dockerfile for Frontend (React) Directory: frontend/

Stage 1: Build

FROM node:18-alpine as build

WORKDIR /app **COPY** package.json ./

RUN npm install **COPY** . .

RUN npm run build

Stage 2: Serve

FROM nginx:alpine

COPY --from=build /app/build /usr/share/nginx/html

EXPOSE 80 **CMD** ["nginx", "-g", "daemon off;"]

Run frontend Container

➤ Build the image:

- **cd frontend**
- `docker build -t frontend:latest`

➤ run image

- `docker run -d -p 80:80 frontend:latest`

2. Dockerfile for Backend (Node.js + Express)

Directory: backend/

FROM node:18-alpine

WORKDIR /app

COPY package.json ./

RUN npm install **COPY** . .

EXPOSE 8080

CMD ["node", "server.js"]

➤ Run backend Container

➤ Build the image:

- **cd backend**
- `docker build -t backend:latest`

➤ run image

- `docker run -d -p 8080:8080 backend:latest`

3. Database: MySQL Container

➤ Environment variables:

- **MYSQL_ROOT_PASSWORD**
- **MYSQL_DATABASE**

Run MySQL Container:- `docker run -d --name my-db --network my-app-net -e MYSQL_ROOT_PASSWORD=root -e MYSQL_DATABASE=mydb -p 3306:3306 --mysql:8`

BREIFING OF DOCKER COMPOSE

Introduction to Docker Compose

What is Docker Compose?

Docker Compose is a tool used to define and manage multi-container Docker applications. It allows developers to describe an application's services, networks, and volumes in a single YAML file (docker-compose.yml) and run them with a single command.

Key Features

- Defines services in a YAML file for clarity and reusability.
- Supports multi-container deployment with a single command.
- Simplifies application setup during development and testing.
- Manages networks and volumes automatically.

Deploying a Three-Tier Application Using Docker Compose

Folder Structure project-root/

├── docker-compose.yml

├── frontend/

│ └── Dockerfile

├── backend/

│ └── Dockerfile

▪ How to Deploy

Step 1: Build and Start Containers

- `docker-compose up --build`

Step 2: Stop and Remove All Services

- `docker-compose down`

➤ Benefits of Using Docker Compose

- Manages multi-container setups with fewer manual steps.
- Ensures consistent local development environments.
- Easier to maintain and version using source control.
- Allows defining networks and volumes declaratively